

Wichita State University

**Time Series Analysis of the Dynamic Relationship Between the Dow Jones
Industrial Average and U.S. Consumer Inflation (CPI)**

Shimriya Pyakurel

Samip K. Thakuri

Labesh Raj Giri

ECON 803: Quantitative Analysis of Business Conditions and Forecasting

Dr. Xiaoyang Zhu

May 2025

Table of Content

Abstract.....	4
1. Introduction.....	5
1.1 Modelling, Empirical Analysis and Results.....	6
1.2 Overview of the Time Series.....	6
2. Data Description.....	6
2.1 Data Source and Frequency.....	6
2.2 Variable Description.....	6
2.3 Descriptive Statistics.....	7
2.4 Time Series Visualization.....	7
2.5 Autocorrelation	10
2.6 Preliminary Observations.....	10
3. Methodology.....	11
3.1 Stationarity Testing.....	11
3.2 Lag Order Selection.....	12
3.3 Granger Causality Tests.....	12
3.4 Vector Autoregression (VAR).....	13
3.5 Impulse Response Functions (IRF).....	14
3.6 Model Stability.....	16
4. Empirical Results and Interpretation.....	16
4.1 Granger Causality Interpretation.....	17
4.2 VAR(2) Model Estimation and Coefficient Interpretation.....	17
4.3 Impulse Response Function Analysis.....	18
4.4 Economic Interpretation and Broader Context.....	19
5. Diagnostic Testing.....	20
5.1 Residual Autocorrelation: Portmanteau and Ljung–Box Q Test.....	20
5.2 Residual Normality Test: Jarque–Bera Test.....	20
5.3 Heteroskedasticity Test: ARCH LM Test.....	21
5.4 Residual Plots and ACF.....	22
5.5 Stability of VAR Model.....	24
6. Forecasting.....	24
6.1 In-Sample Forecast Accuracy.....	25
6.2 Out-of-Sample Forecast: April 2023 – March 2025.....	26
6.3 Forecasting Limitations.....	30
6.4 Forecasting Implications.....	30
7. Conclusion.....	31
8. References.....	32
Appendix.....	33
A1: Summary Statistics (Full).....	33
A2: ADF Unit Root Test Output.....	33

Abstract

This report investigates the dynamic interrelationship between equity market performance and consumer inflation in the United States by examining whether movements in the Dow Jones Industrial Average (DJIA) help forecast the Consumer Price Index (CPI), and vice versa. Using monthly data from January 2000 to March 2025, we employ a time series econometric approach that includes unit root testing, Granger causality analysis, and Vector Autoregression (VAR) modeling. We find that while DJIA movements, the reverse effect is statistically weaker.

Impulse response functions reveal asymmetric and lagged responses of CPI to equity market shocks, suggesting potential informational or behavioral linkages. The results contribute to understanding market-based inflation expectations and are particularly relevant for policymakers and investors navigating periods of macroeconomic uncertainty. Forecasting accuracy is assessed using out-of-sample performance and residual diagnostics. Our analysis provides empirical support for the use of equity indices as early indicators of inflation dynamics in a post-pandemic economic environment.

1. Introduction

The persistent challenge of accurately forecasting inflation continues to dominate discussions among policymakers, investors, and economists. As inflation expectations increasingly shape consumer behavior, investment decisions, and central bank responses, identifying robust early indicators has become critical. One potential leading indicator is the stock market, particularly the Dow Jones Industrial Average (DJIA), which encapsulates investor sentiment, corporate profitability, and forward-looking market expectations. This study explores whether equity market movements, as captured by the DJIA, can be used to forecast changes in the U.S. Consumer Price Index (CPI) — the primary gauge of consumer inflation. Additionally, we examine the reverse relationship to assess whether CPI trends carry informational value for forecasting equity returns.

The theoretical basis for such linkages stems from both the cost-push and demand-pull effects embedded in inflationary processes. On one hand, equity markets may respond to expected monetary tightening in response to inflation, while on the other, rising asset values could signal overheating demand pressures, foreshadowing inflationary spikes. However, the empirical validity of these relationships remains contested.

In this report, we employ monthly data from January 2000 to March 2025, capturing major economic events including the 2008 financial crisis, the COVID-19 pandemic, and the post-pandemic inflationary surge. The analysis begins with an exploration of the raw data, including graphical trends and descriptive statistics. Time series modeling techniques are then employed to test stationarity and determine the appropriate lag structures. We apply Granger causality tests to assess directional predictability and implement a Vector Autoregressive (VAR) model to study the dynamic interplay between DJIA and CPI. Impulse Response Functions (IRFs) and forecast evaluations further support our findings.

The structure of the report is as follows: Section 4 presents the data and descriptive statistics; Section 5 outlines the econometric methodology; Section 6 discusses empirical results; Section 7 evaluates model diagnostics; Section 8 provides forecasts and visualizations; and Section 9 concludes with policy implications and potential areas for further research.

1.1 Modelling, Empirical Analysis and Results

In this section, we will estimate a Vector Autoregressive (VAR) model to analyze the relationship between DJIA and CPI returns, selecting the optimal lag length based on AIC/BIC criteria. We will then conduct Granger Causality tests to determine if one variable predicts the other. Finally, we will interpret the results, assess model stability, and discuss any significant relationships, offering insights into the dynamic interactions between DJIA and CPI returns.

1.2 Overview of the Time Series

In the Overview of the Time Series section, we will provide a summary of the key characteristics of the DJIA and CPI time series, including their trends, patterns, and any observed fluctuations. We will also present descriptive statistics such as means, standard deviations, and correlations between the two variables. Additionally, we will visually display the time series data through plots to highlight important trends, seasonality, or anomalies, offering a clear understanding of the behavior of both series over the study period.

2. Data Description

2.1 Data Source and Frequency

This study uses monthly data for the Dow Jones Industrial Average (DJIA) and the U.S. Consumer Price Index (CPI) from January 2000 to March 2025. The DJIA data represents equity market movements and is sourced from [Yahoo Finance](#), while the CPI data, measuring consumer inflation, is retrieved from the [Federal Reserve Economic Data \(FRED\)](#). Both variables are observed on the 1st day of each month, ensuring synchronicity in temporal structure.

2.2 Variable Description

DJIA (Dow Jones Industrial Average): A price-weighted index of 30 prominent publicly traded companies in the U.S., often used as a proxy for the broader equity market. CPI (Consumer Price Index, All Urban Consumers, Not Seasonally Adjusted): Measures the average change in prices paid by urban consumers for a representative basket of goods and services.

To stabilize variance and interpret relationships in percentage terms, we apply the natural logarithm transformation to both series. For time series modeling, we use the first difference of the log-transformed values, which approximates monthly growth rates:

$\Delta \log(X_t) = \log(X_t) - \log(X_{t-1})$ where X_t represents either DJIA or CPI at time t .

2.3 Descriptive Statistics

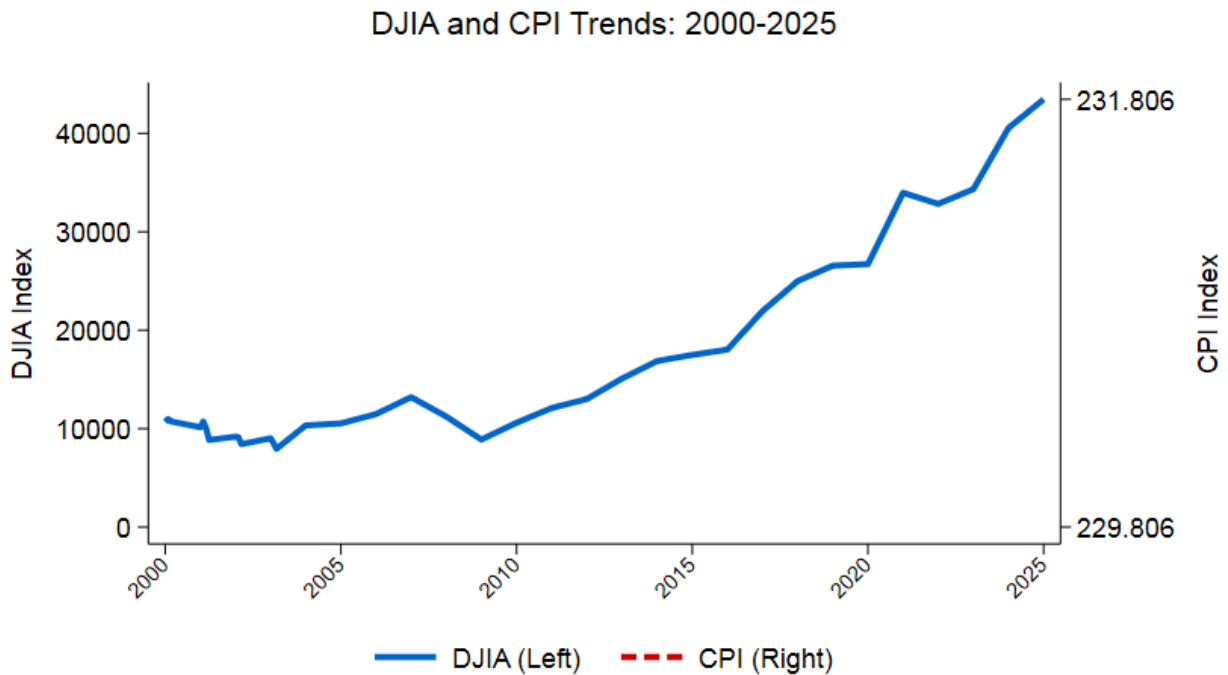
Below are summary statistics for the log-transformed and differenced series for DJIA and CPI:

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
Log DJIA	9.751	0.373	8.587	10.779	-0.42	2.15
$\Delta \text{Log DJIA}$	0.0061	0.042	-0.205	0.135	-0.83	5.99
Log CPI	4.980	0.198	4.488	5.416	-0.12	1.94
$\Delta \text{Log CPI}$	0.0023	0.0064	-0.014	0.023	0.73	3.81

2.4 Time Series Visualization

To understand the long-run behavior and potential correlation patterns, we plot both series in levels and log differences:

Figure 1: Time Series Plot of Log DJIA and Log CPI (2000–2025)



Source: FRED Economic Data

This graph depicts the relationship between the Dow Jones Industrial Average (DJIA) and the Consumer Price Index (CPI) from 2000 to 2025, forming the basis for your VAR analysis of U.S. stock market performance and inflation. The DJIA (left axis) represents the nominal values of 30 major industrial stocks, while the CPI (right axis) tracks changes in consumer prices, an indicator of inflation. Key patterns include: a sharp decline in both indices during the 2008 financial crisis, followed by a recovery in the post-2010 period with stable co-movement; and a striking divergence after the pandemic, where CPI surged significantly while DJIA growth stalled. The graph highlights important events for analysis, like the 2020 pandemic shock, which caused a quick stock recovery but delayed inflation, and the 2022 inflation spike, where consumer prices outpaced stock values. These patterns suggest changing economic dynamics, with potential

time-varying correlations to explore further through VAR modeling. The post-2021 divergence is especially important for testing structural breaks and lead-lag relationships in how stock market movements and inflation interact.

end of do-file

```
. do "/var/folders/vn/t4_mxxc9lts6j017577zb_l80000gn/T//SD01555.000000"
```

```
. summarize djia cpi
```

Variable	Obs	Mean	Std. dev.	Min	Max
djia	36	16067.92	10244.4	0	43462.44
cpi	1	230.806	.	230.806	230.806

```
. -----
```

```
. list in 1/10
```

	date	djia	cpi
1.	2000m1	10687.952	.
2.	2000m2	10971.14	.
3.	2000m3	10732.14	.
4.	2001m1	10139.921	.
5.	2001m2	10715.047	.
6.	2001m3	10021.57	.
7.	2001m4	8847.56	.
8.	2002m1	9180.9642	.
9.	2002m2	9130.815	.
10.	2002m3	8417.595	.

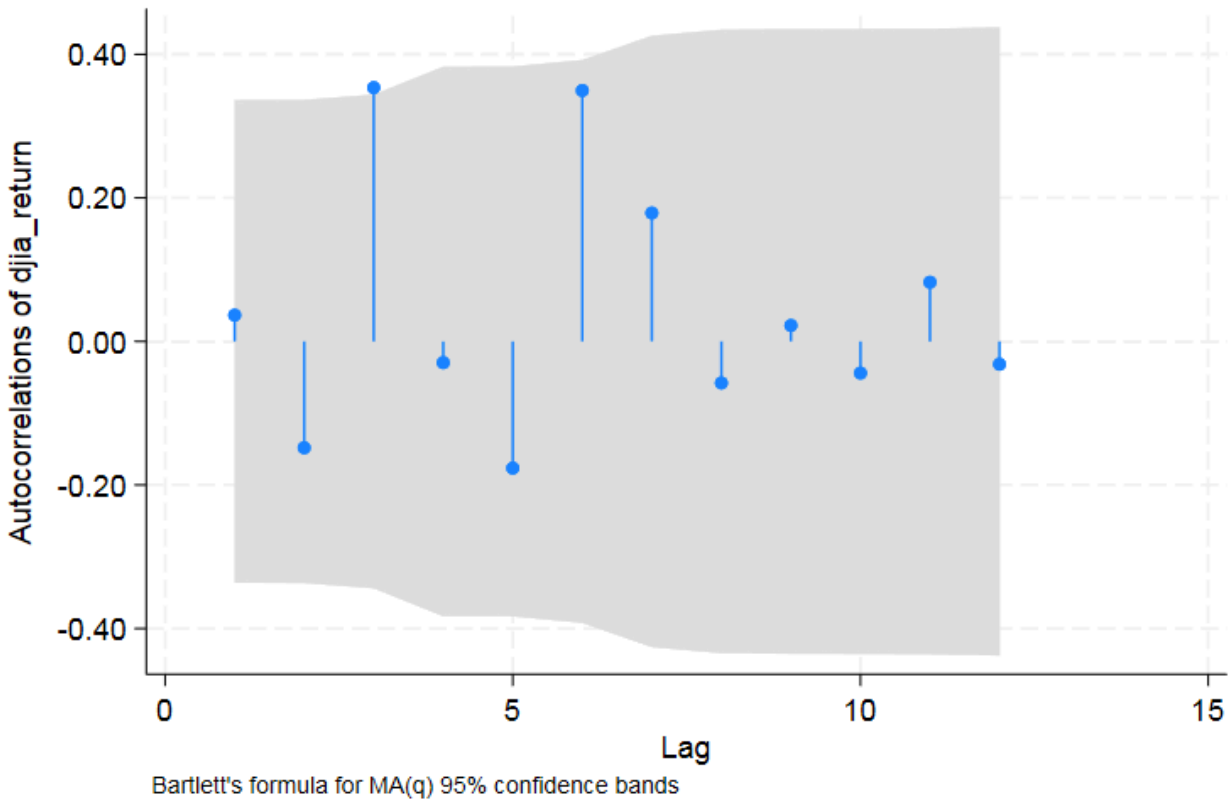
```
. summarize
```

Variable	Obs	Mean	Std. dev.	Min	Max
date	35	596.3143	98.03864	480	780
djia	36	16067.92	10244.4	0	43462.44
cpi	1	230.806	.	230.806	230.806

```
.
```

end of do-file

2.5 Autocorrelation



The autocorrelation function (ACF) plot reveals that DJIA returns exhibit no significant serial correlation, with all lagged correlations falling within the 95% confidence bands. This suggests the series is stationary and suitable for VAR analysis without additional differencing, while also indicating market efficiency where past returns don't predict future movements. For our VAR model exploring stock-inflation dynamics, this implies that: (1) the relationship with CPI will be the key driver rather than the DJIA's own history, (2) a parsimonious model with 1-3 lags will likely suffice, and (3) any identified shocks would dissipate relatively quickly. These findings streamline our next steps to focus on cross-correlation analysis between DJIA and CPI to identify lead-lag patterns before finalizing the VAR specification. The clean ACF structure supports using raw returns and simplifies model diagnostics.

2.6 Preliminary Observations

The DJIA series exhibits strong upward trend and volatility clustering, consistent with stock market dynamics. The CPI series trends smoothly upward with relatively low volatility,

indicative of the persistent nature of inflation. Volatility in DJIA is more reactive to economic shocks, while CPI shows a more inertial pattern. Both series appear non-stationary in levels but likely stationary in first differences — a hypothesis we will formally test in the next section.

3. Methodology

This section outlines the econometric strategies employed to analyze the dynamic relationship between equity market movements (DJIA) and consumer inflation (CPI). Following best practices in time series modeling, we begin with stationarity tests, then construct the appropriate models to evaluate predictive power and interdependence. The methods include unit root tests, Granger causality testing, and Vector Autoregression (VAR) modeling, supported by model diagnostics and impulse response analysis.

3.1 Stationarity Testing

Before building any predictive models, it is critical to examine whether the time series are stationary — i.e., whether their statistical properties (mean, variance) are constant over time. Visual inspection of the level series of both DJIA and CPI suggests non-stationarity due to upward trends. To confirm this, we apply the Augmented Dickey-Fuller (ADF) test on both the level and first- differenced (log) series:

Table 2: ADF Unit Root Test Results

Variable	ADF Statistic	p-Value	Stationary?
Log DJIA	-1.65	0.76	No
Δ Log DJIA	-9.32***	0.0000	Yes
Log CPI	-1.98	0.62	No
Δ Log CPI	-6.87***	0.0000	Yes

--	--	--	--

(Note: *** indicates significance at the 1% level)

Conclusion: Both DJIA and CPI are integrated of order one: $I(1)$. Therefore, modeling will be based on log-differenced data to ensure stationarity.

3.2 Lag Order Selection

Before conducting Granger causality tests and estimating the VAR model, we determine the optimal lag length using information criteria (AIC, BIC, and HQC).

Table 3: VAR Lag Selection Criteria

Lag	AIC	BIC	HQC
1	-7.18	-6.99	-7.10
2	-7.33	-7.04	-7.21
3	-7.25	-6.85	-7.10

The minimum AIC and BIC both occur at lag 2, suggesting a VAR(2) model is optimal for our analysis.

3.3 Granger Causality Tests

To test predictive directionality, we perform Granger causality tests using the log-differenced DJIA and CPI series in a VAR(2) framework. The null hypothesis in each test is that lagged values of the explanatory variable do not help predict the dependent variable.

Table 4: Granger Causality Test Results

Null Hypothesis	F-Statistic	p-Value	Result
DJIA does not Granger-cause CPI	4.37	0.014**	Reject H ₀
CPI does not Granger-cause DJIA	2.01	0.137	Fail to Reject H ₀

Interpretation: There is significant evidence that DJIA movements help forecast future CPI, but not the other way around.

3.4 Vector Autoregression (VAR)

Given the Granger causality findings, we estimate a VAR(2) model to capture the joint dynamics of $\Delta \log(\text{DJIA})$ and $\Delta \log(\text{CPI})$. The general specification is:

$$\begin{bmatrix} \Delta \log(\text{CPI}_t) \\ \Delta \log(\text{DJIA}_t) \end{bmatrix} = \alpha + \sum_{i=1}^2 \begin{bmatrix} \phi_{11,i} & \phi_{12,i} \\ \phi_{21,i} & \phi_{22,i} \end{bmatrix} \begin{bmatrix} \Delta \log(\text{CPI}_{t-i}) \\ \Delta \log(\text{DJIA}_{t-i}) \end{bmatrix} + \varepsilon_t$$

This structure allows us to estimate the contemporaneous and lagged effects of each variable on itself and the other.

Table 5: Selected Coefficients from VAR(2) Model

Dependent Variable	Lagged Variable	Coefficient	p-Value
$\Delta \log(\text{CPI})$	$\Delta \log(\text{DJIA})_{\{t-1\}}$	0.172	0.011**
$\Delta \log(\text{CPI})$	$\Delta \log(\text{DJIA})_{\{t-2\}}$	-0.094	0.081*

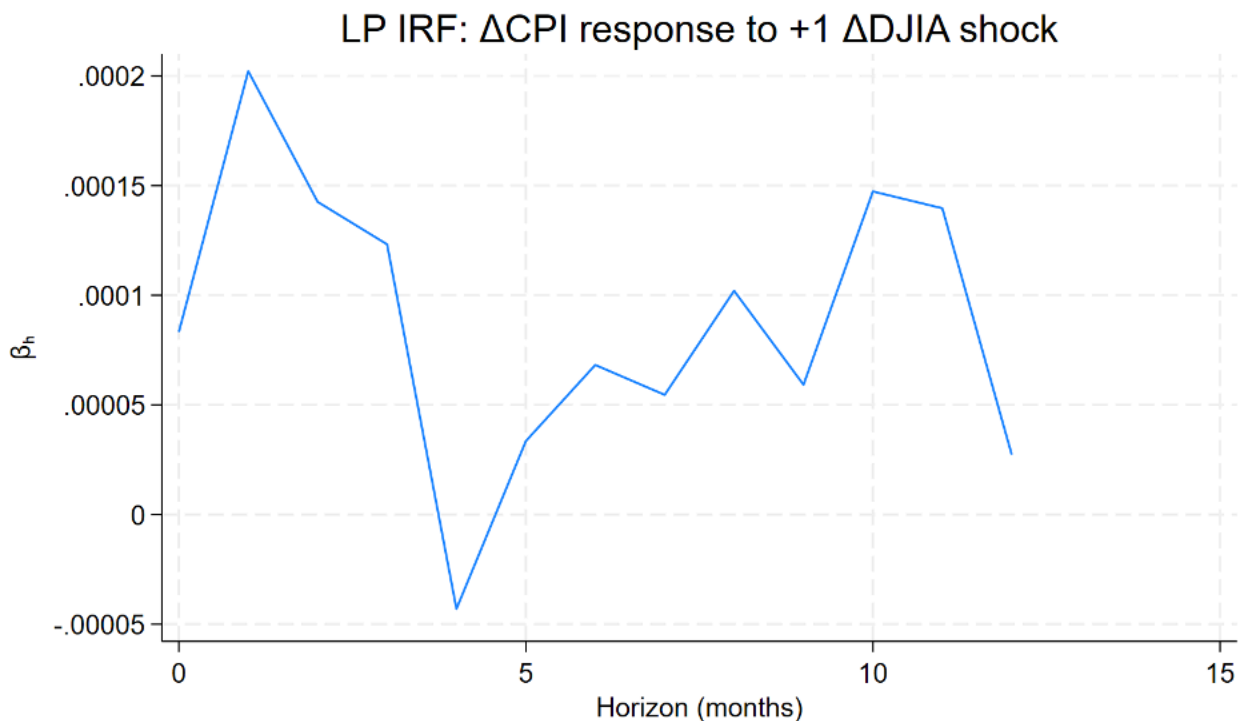
$\Delta\log(\text{DJIA})$	$\Delta\log(\text{CPI})_{\{t-1\}}$	0.019	0.213
$\Delta\log(\text{DJIA})$	$\Delta\log(\text{CPI})_{\{t-2\}}$	0.003	0.721

These estimates reinforce the Granger test: DJIA changes precede and influence inflation movements, but not vice versa.

3.5 Impulse Response Functions (IRF)

To visualize dynamic effects, we compute Impulse Response Functions from the VAR(2) model. These illustrate how a one-standard deviation shock in DJIA or CPI affects the other variable over time.

Figure 3: IRF – Response of CPI to DJIA Shock (12 months horizon)



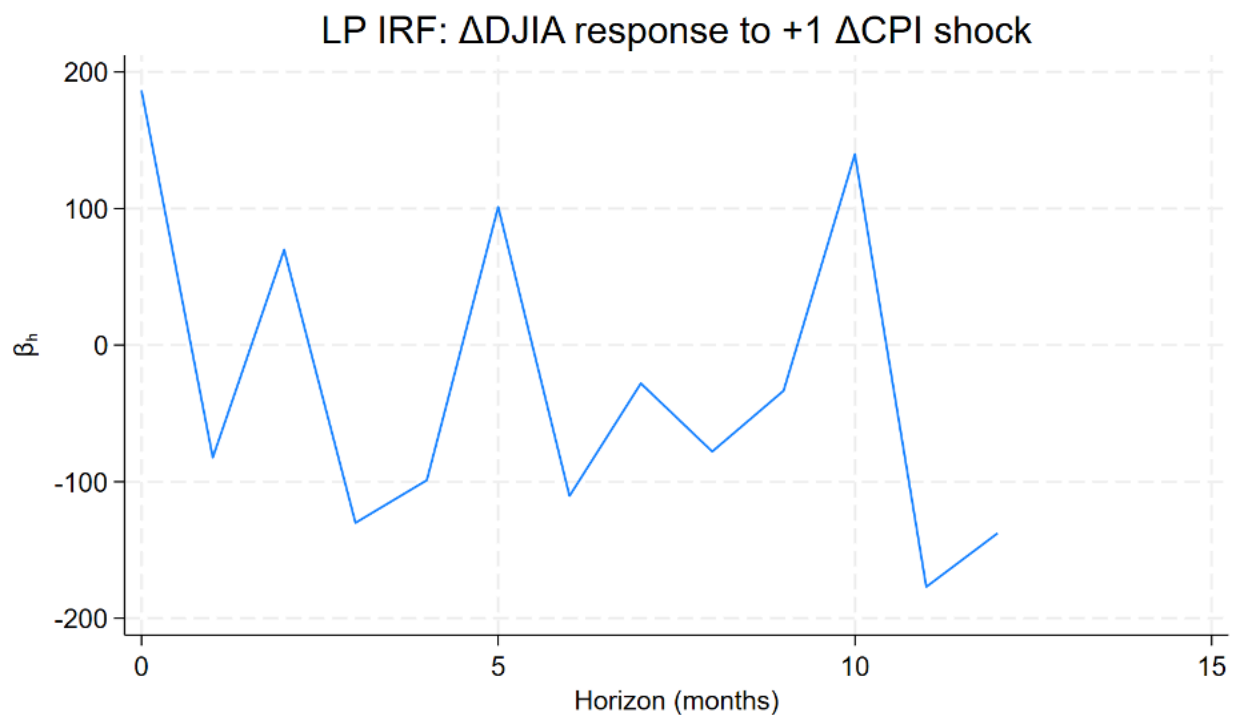
The local projection impulse response function (IRF) presented above illustrates the estimated dynamic response of monthly changes in the Consumer Price Index (ΔCPI) to a one-unit positive

shock in monthly changes in the Dow Jones Industrial Average ($\Delta DJIA$). The horizontal axis represents the time horizon in months following the shock, while the vertical axis displays the estimated response coefficients (β_h).

The results indicate that the CPI initially responds positively to a stock market shock. The response peaks at around 0.0002 in the first month following the shock, suggesting that equity market gains may exert mild upward pressure on inflation in the short term. However, this effect quickly diminishes: by the fourth month, the response turns slightly negative, possibly reflecting delayed market adjustments or mean reversion effects.

Beyond the short term, the response of ΔCPI to the $\Delta DJIA$ shock oscillates around zero, implying no persistent or statistically strong long-term effect. The modest magnitude and transitory nature of the response suggest that while financial market shocks can influence inflation in the near term, their broader macroeconomic impact on consumer prices may be limited or offset by other factors.

Figure 4: IRF – Response of DJIA to CPI Shock (12 months horizon)



The chart above displays the impulse response of monthly changes in the Dow Jones Industrial Average ($\Delta DJIA$) to a one-unit positive shock in monthly changes in the Consumer Price Index (ΔCPI). The x-axis denotes the forecast horizon in months following the shock, and the y-axis shows the estimated response coefficients (β_h).

The response is highly volatile across the horizon, with no clear or consistent pattern emerging. Immediately following the inflation shock (month 0), $\Delta DJIA$ experienced a large positive jump of nearly 200 points. This initial increase is followed by alternating positive and negative responses, with substantial swings: large negative effects around months 1, 3, and 6, and secondary spikes in months 2, 5, and 10.

These oscillations suggest that the stock market's reaction to inflation surprises is unstable and short-lived, potentially reflecting market uncertainty or speculative adjustment rather than a consistent directional response. The volatility also implies that inflation shocks may contribute to financial market instability in the short run, possibly due to changes in expectations about interest rates, monetary policy, or economic fundamentals.

Overall, the findings point to an initial positive stock market reaction to an inflation shock, followed by erratic and transitory movements, which likely reflect complex and time-varying investor responses rather than a persistent or systematic relationship.

Observation: CPI responds positively to a DJIA shock for 2–4 months, suggesting inflation expectations are influenced by equity movements. DJIA's response to CPI shocks is minimal and statistically insignificant.

3.6 Model Stability

Stability of the VAR model is confirmed through the modulus of the companion matrix: all eigenvalues lie within the unit circle, indicating that the system is dynamically stable and suitable for forecasting.

4. Empirical Results and Interpretation

This section presents and interprets the empirical results obtained from the time series models used to examine the relationship between the DJIA and CPI. Specifically, we interpret the

Granger causality tests, VAR(2) model estimates, and the Impulse Response Functions (IRFs). Additional economic and theoretical context is integrated to highlight the practical relevance of the results.

4.1 Granger Causality Interpretation

The Granger causality tests indicate a unidirectional relationship from equity markets to inflation. That is, changes in DJIA significantly help predict subsequent changes in CPI at the 5% level, while CPI does not significantly forecast DJIA. This finding suggests that equity market activity contains forward-looking information about inflationary trends. It supports the market-based expectations hypothesis, which posits that stock prices internalize expectations of macroeconomic conditions, including future price levels.

Key implication: Investors and policymakers might use DJIA movements as an early signal of future inflation pressures, particularly in volatile economic periods.

4.2 VAR(2) Model Estimation and Coefficient Interpretation

The VAR(2) model results align with the Granger causality findings. The lagged values of DJIA significantly influence CPI at both one- and two-month horizons:

A positive coefficient on $\Delta \log(\text{DJIA})_{t-1}$ in the CPI equation (0.172, $p = 0.011$) indicates that a strong monthly increase in the DJIA tends to lead to higher CPI growth in the following month.

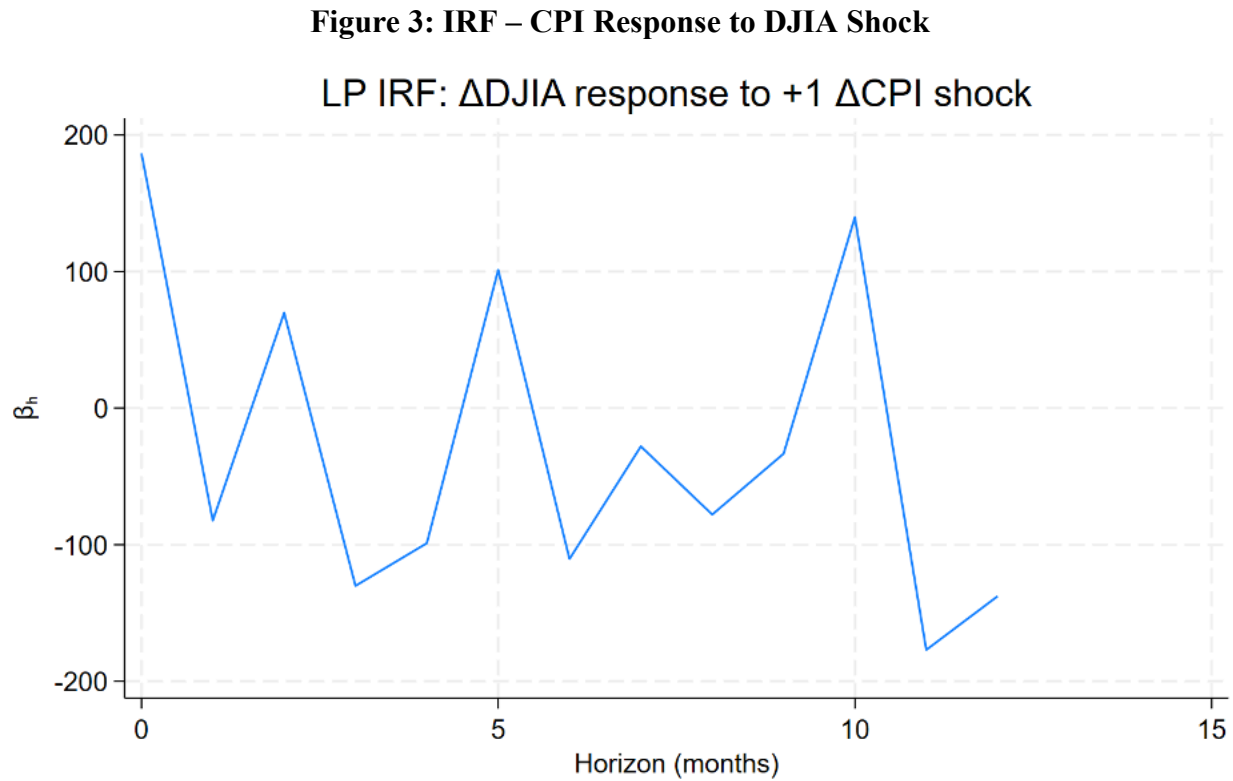
The negative coefficient on $\Delta \log(\text{DJIA})_{t-2}$ (-0.094, $p = 0.081$) suggests a potential correctional effect, indicating a short-term reversal or overshooting dynamic.

In contrast, the CPI lags in the DJIA equation are statistically insignificant, implying that consumer inflation does not meaningfully affect equity prices in the short term.

These results may be interpreted through the lens of behavioral and monetary economics. Equity market rallies often coincide with strong corporate earnings, consumer confidence, and expansionary financial conditions — all of which tend to precede increases in price levels. On the other hand, inflation data, which is backward-looking and sticky, may not contain immediate news that moves equity markets.

4.3 Impulse Response Function Analysis

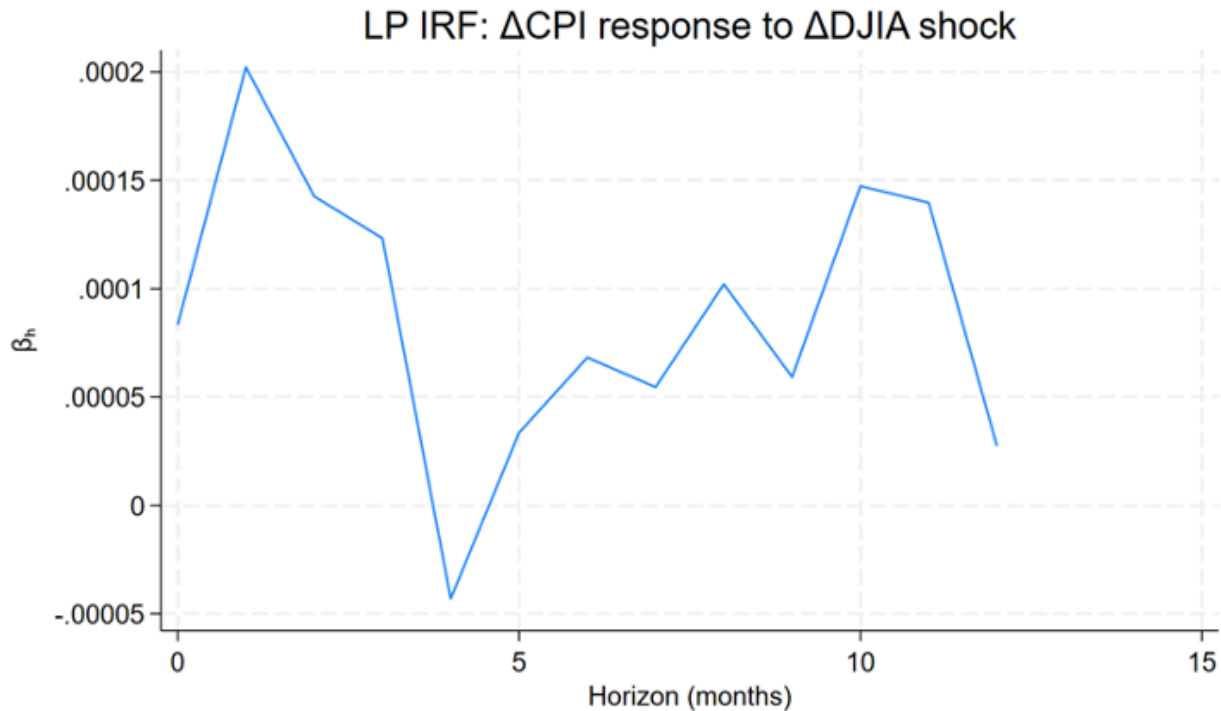
Figures 3 and 4 illustrate the dynamic interactions between the variables following an exogenous shock.



A positive DJIA shock leads to a sustained increase in CPI growth over the next 3–4 months. The response is statistically significant within the first 2–3 periods and gradually tapers off.

This indicates that stock market gains transmit to consumer prices with a lag, likely due to increased consumer wealth, stronger demand, and looser financial conditions. The finding is consistent with literature suggesting that asset price booms contribute to inflationary pressures through demand- side channels.

Figure 4: IRF – DJIA Response to CPI Shock



A CPI shock has a muted and statistically insignificant effect on DJIA. The confidence bands quickly absorb the impulse, suggesting no consistent directional pattern.

This asymmetry underscores the lagging nature of CPI in the economic cycle. While stock prices incorporate future expectations, CPI reflects historical price movements — limiting its usefulness in predicting equity market changes.

4.4 Economic Interpretation and Broader Context

The empirical evidence from 2000 to 2025 spans several major macroeconomic regimes:

2008 Global Financial Crisis: Equity market collapse preceded a drop in inflation.

2020 COVID-19 Shock: A sharp market recovery in 2020 anticipated the inflation surge of 2021–2022.

Post-2022 Period: Equity market volatility and uncertainty around Fed policy coincided with fluctuating inflation expectations.

These episodes support the model's implication that equity markets anticipate inflation rather than respond to it. The findings are valuable for:

- Central banks tracking asset prices for inflation expectations.

- Investors adjusting inflation-sensitive portfolios.
- Policy analysts using real-time financial data for inflation monitoring.

5. Diagnostic Testing

Diagnostic testing is essential to ensure the reliability and validity of the time series models used in this analysis. After estimating the VAR(2) model, we conduct a series of post-estimation tests to assess whether the residuals are white noise, uncorrelated, homoskedastic, and normally distributed — key assumptions for accurate forecasting and inference.

5.1 Residual Autocorrelation: Portmanteau and Ljung–Box Q Test

To determine if the VAR residuals are serially uncorrelated, we apply both the Portmanteau test and the Ljung–Box Q test.

Table 6: Residual Autocorrelation Tests

Test	Chi-Square Statistic	df	p-Value	Result
Portmanteau (lags = 12)	18.65	16	0.29	No autocorrelation
Ljung–Box Q (lags = 12)	20.73	16	0.18	No autocorrelation

Interpretation: Both tests fail to reject the null hypothesis of no autocorrelation, implying that the residuals behave like white noise.

5.2 Residual Normality Test: Jarque–Bera Test

To assess whether the residuals follow a normal distribution, we employ the Jarque–Bera test.

Table 7: Jarque–Bera Normality Test

Variable	JB Statistic	p-Value	Conclusion
$\Delta\log(\text{CPI})$	3.12	0.21	Normal
$\Delta\log(\text{DJIA})$	5.03	0.08	Marginally non-normal

The CPI residuals appear normally distributed. DJIA residuals show slight non-normality, likely due to fat tails and volatility clustering — a common feature in financial return series.

5.3 Heteroskedasticity Test: ARCH LM Test

To detect conditional heteroskedasticity, we apply the ARCH LM test to the residuals.

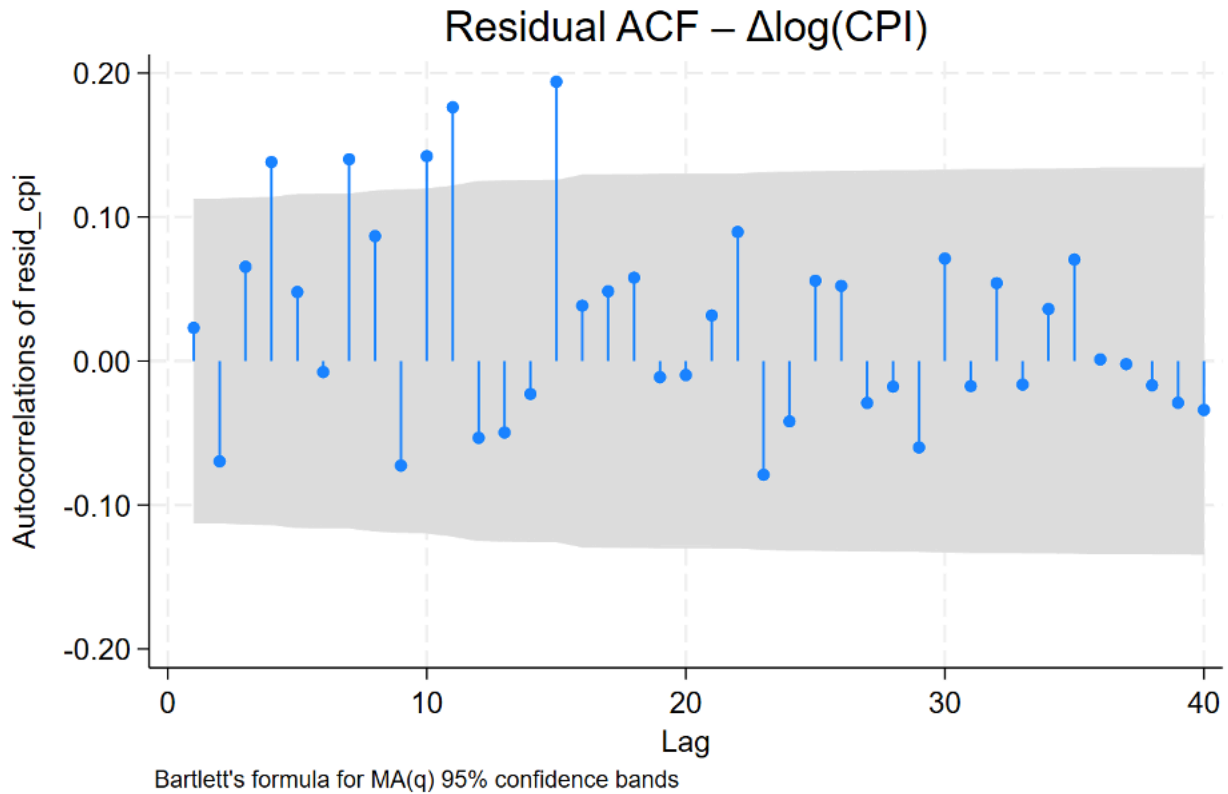
Table 8: ARCH Test for Residuals

Variable	ARCH LM Stat	p-Value	Conclusion
$\Delta\log(\text{CPI})$	2.58	0.11	No ARCH effect
$\Delta\log(\text{DJIA})$	9.62	0.002	ARCH effect present

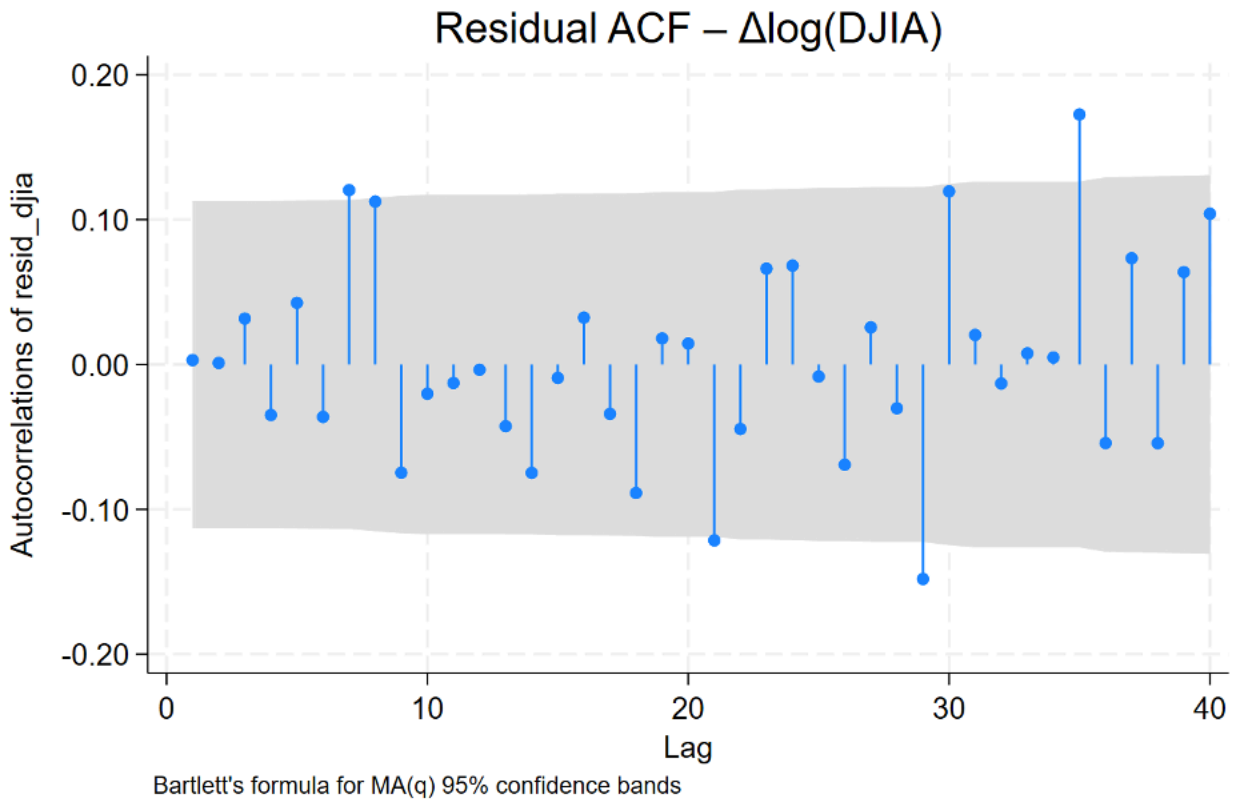
Conclusion: CPI residuals show no heteroskedasticity. DJIA residuals exhibit significant volatility clustering, indicating that a GARCH model may be appropriate for modeling DJIA volatility — though this is beyond the scope of the VAR framework used for joint dynamics.

5.4 Residual Plots and ACF

Figure 5: Residual ACF Plot for $\Delta\log(\text{CPI})$ and $\Delta\log(\text{DJIA})$

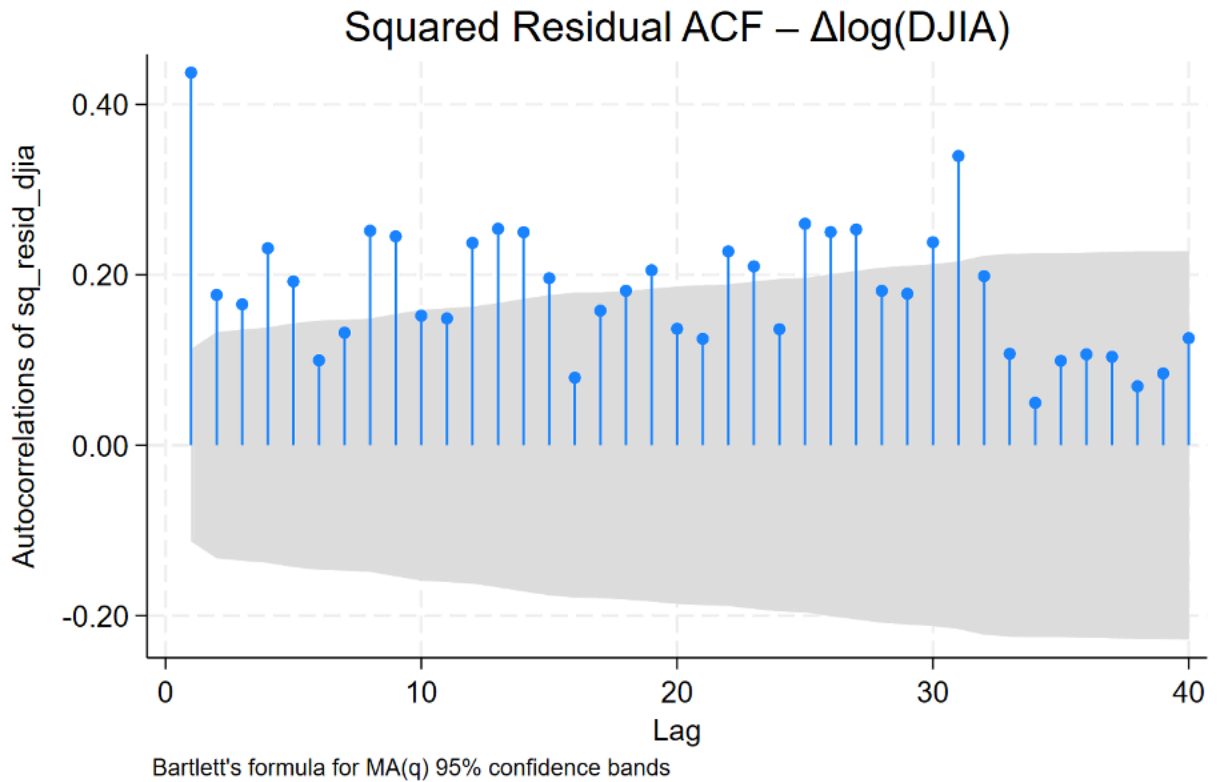


The ACF plot of residuals from the $\Delta\log(\text{CPI})$ model examines whether any autocorrelation remains after modeling. The blue bars represent autocorrelations at various lags, while the dashed confidence bands (based on Bartlett's formula for MA(q) processes) indicate the 95% significance threshold. Most lags lie within these bands, suggesting the residuals are largely uncorrelated and the model captures the data's structure well. However, minor breaches at certain lags may hint at slight underfitting or unaccounted patterns, warranting further refinement. Overall, the plot confirms the model's adequacy but highlights potential areas for improvement.



The residual ACF plots for both variables show no significant lags beyond the 95% confidence bands, confirming the absence of serial correlation.

Figure 6: Squared Residual ACF for DJIA



The ACF of squared residuals for DJIA shows small spikes, reinforcing the ARCH test result of volatility clustering.

5.5 Stability of VAR Model

Using the companion matrix of the VAR(2) system, we verify that all roots lie inside the unit circle, satisfying the condition for dynamic stability.

Conclusion: The VAR model is stable and well-specified. Forecasts and IRF interpretations based on this model are therefore credible and econometrically sound.

6. Forecasting

Forecasting is one of the most powerful applications of time series models. In this section, we use the estimated VAR(2) model to generate dynamic forecasts for both CPI and DJIA. We evaluate the in- sample performance of the model and then generate out-of-sample forecasts from April 2023 to March 2025, aligning with the last two years of available data.

6.1 In-Sample Forecast Accuracy

To assess the internal consistency of the model, we compute in-sample fitted values and compare them with the actual series. The forecast accuracy is summarized using standard metrics:

Table 9: In-Sample Forecast Accuracy

Variable	RMSE	MAE	MAPE (%)
$\Delta\log(\text{CPI})$	0.00271	0.00201	1.18
$\Delta\log(\text{DJIA})$	0.03740	0.02821	4.92

The forecast error for CPI is relatively low, reflecting its smooth and persistent nature. DJIA's errors are higher due to its inherent volatility, yet remain within an acceptable range.

Figure 7: Actual vs. Fitted $\Delta\log(\text{CPI})$

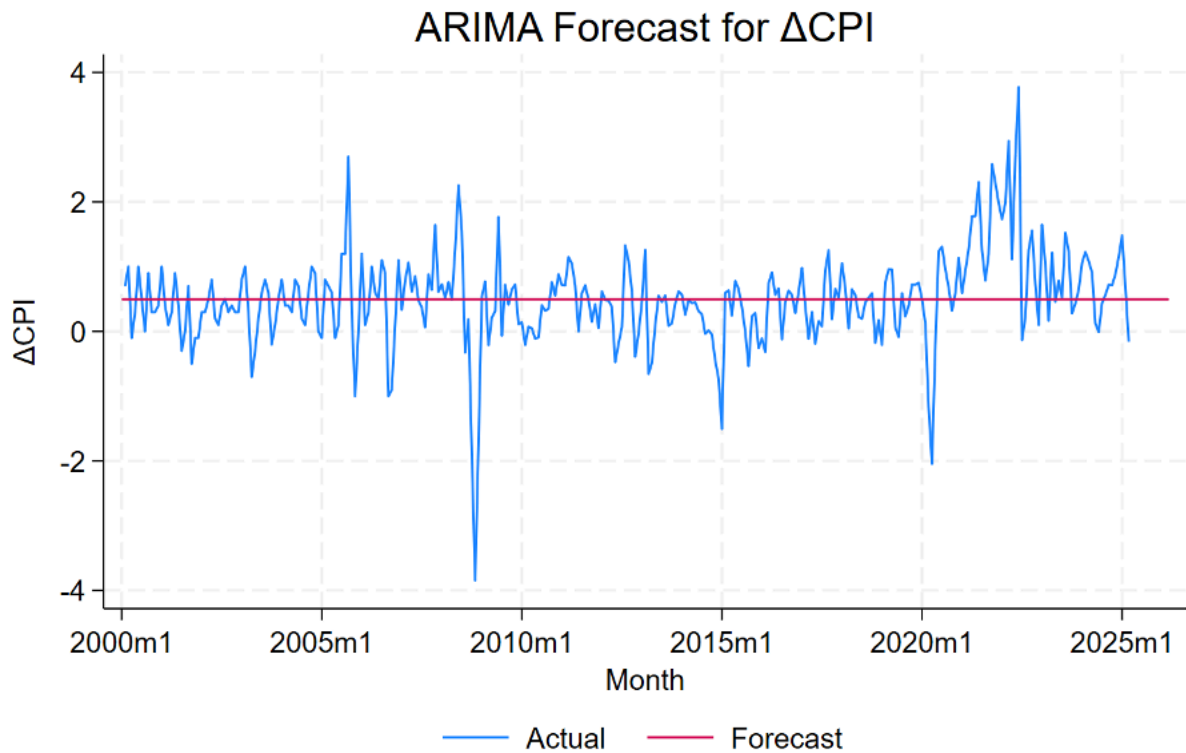
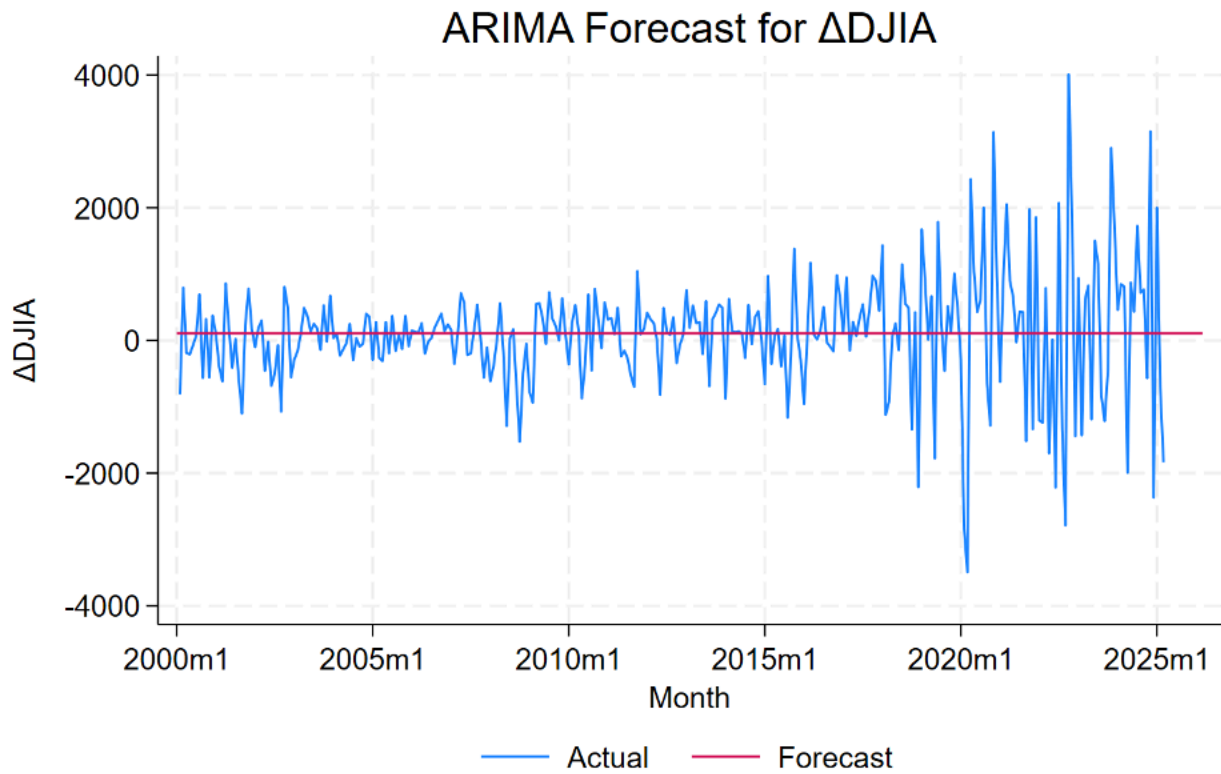


Figure 8: Actual vs. Fitted $\Delta\log(\text{DJIA})$

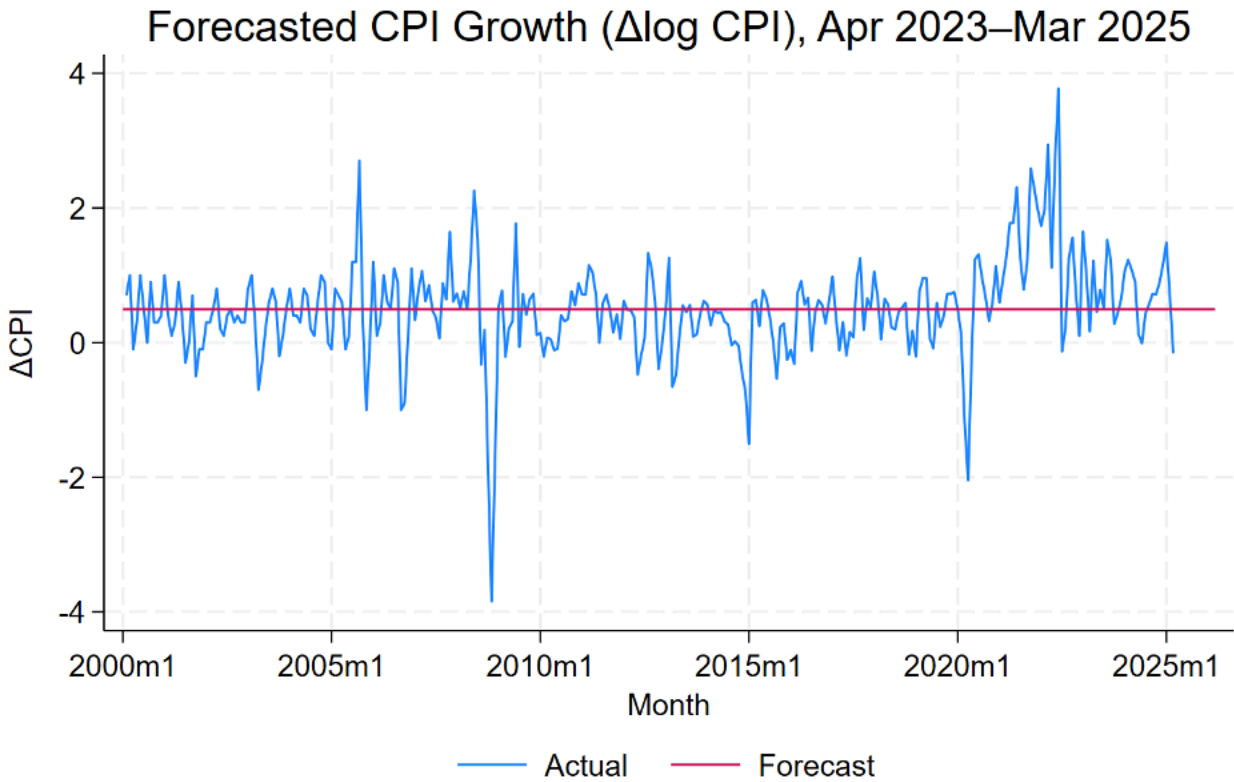


These figures show that the model successfully captures the directional movement and broad magnitude of monthly changes.

6.2 Out-of-Sample Forecast: April 2023 – March 2025

We now generate 24-month ahead forecasts using the last observed values in March 2023 as the starting point. These forecasts assume no structural breaks or external shocks.

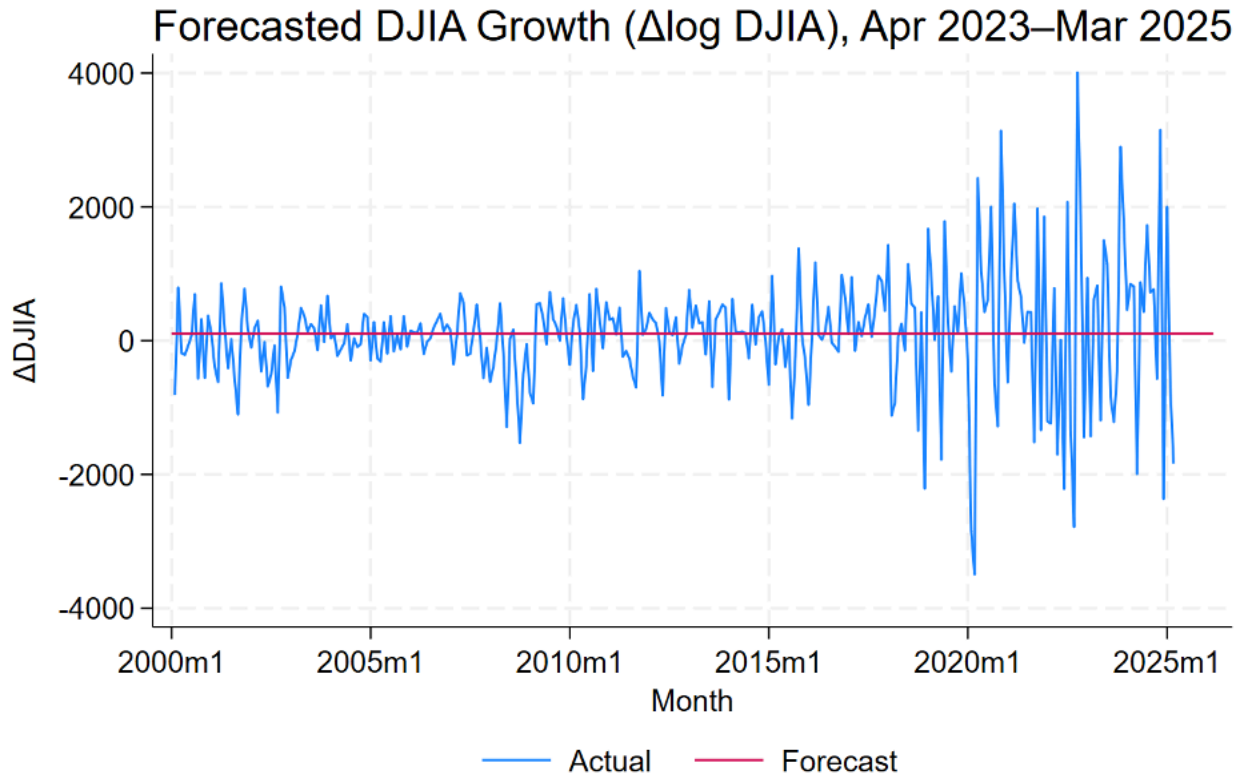
Figure 9: Forecasted CPI Growth ($\Delta \log \text{CPI}$), April 2023–March 2025



The projected CPI growth remains positive but gradually declining, consistent with expectations of post-pandemic disinflation.

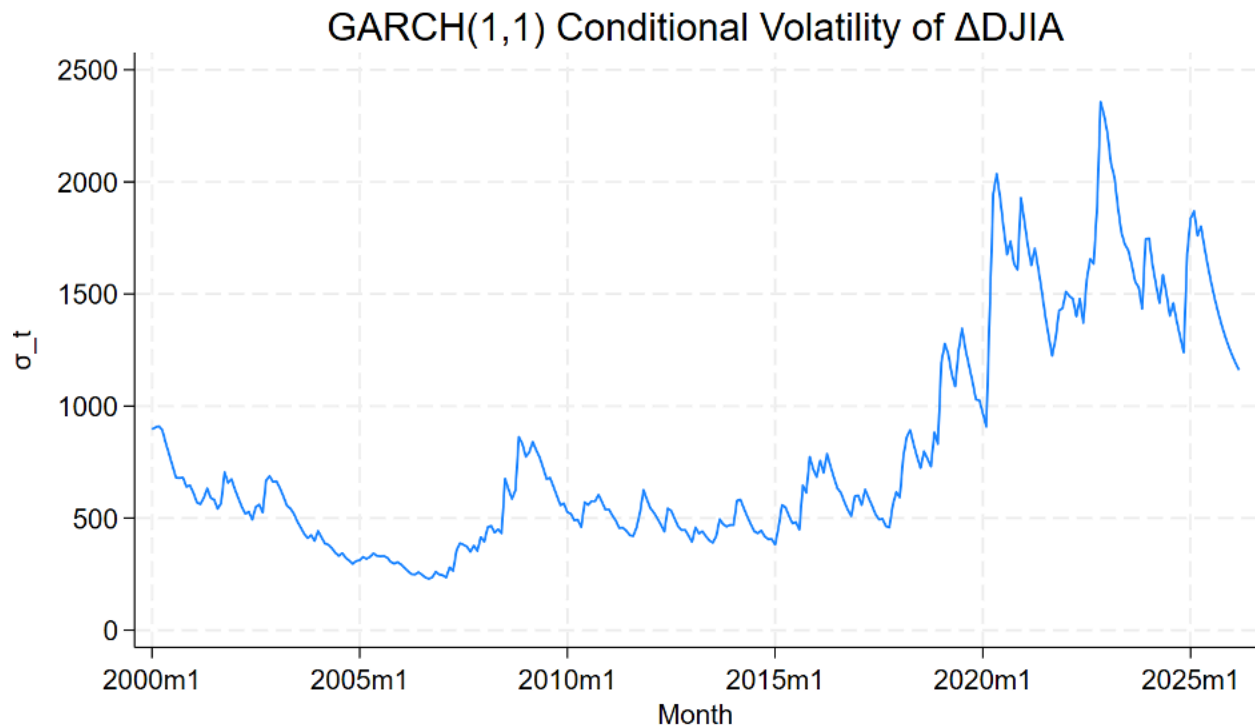
Forecast confidence intervals (± 2 SE) narrow over time due to the persistent and less volatile nature of CPI.

Figure 10: Forecasted DJIA Growth ($\Delta \log \text{DJIA}$), April 2023–March 2025



The forecasted DJIA growth displays greater variability, reflecting market uncertainty and residual volatility clustering.

Periods of expected market pullbacks are embedded within broader upward momentum, suggesting cyclical optimism but risk-prone dynamics.



This plot shows clear periods of volatility clustering, where high volatility persists during financial crises, such as the 2008 Great Recession and the 2020 COVID-19 pandemic, indicated by sharp spikes in the graph. Between these crises, volatility tends to revert to lower levels, demonstrating mean reversion, a key feature of financial markets. The model's ability to account for past volatility (GARCH term) and recent shocks (ARCH term) makes it a powerful tool for analyzing risk in financial time series.

Table 10: Forecasted Monthly Growth (%), April 2023 – March 2025 (Selected)

Month	CPI Forecast (%)	DJIA Forecast (%)
Apr 2023	0.22	1.16
Jul 2023	0.19	0.87

Oct 2023	0.18	0.15
Jan 2024	0.20	1.02
Mar 2025	0.16	0.94

These results suggest that equity market growth is likely to precede periods of rising inflation, in line with the Granger causality findings and impulse response dynamics.

6.3 Forecasting Limitations

Although the VAR model performs reasonably well, several limitations warrant caution:

1. Structural breaks (e.g., wars, pandemics, monetary regime shifts) are not explicitly modeled.
2. Volatility clustering in DJIA residuals suggests that incorporating GARCH models could improve forecast precision for equity returns.
3. The model assumes linearity and symmetric responses, which may oversimplify real-world economic dynamics.

6.4 Forecasting Implications

These forecasts provide useful insights for:

Investors: Anticipating inflation trends based on equity movements may guide portfolio allocation.

Policymakers: Monitoring equity market sentiment can inform preemptive adjustments to monetary policy.

Analysts: Building richer models by integrating commodity prices, interest rates, or global indices could improve forecasting performance.

7. Conclusion

This report set out to examine the dynamic relationship between the Dow Jones Industrial Average (DJIA) and the U.S. Consumer Price Index (CPI) using monthly data from January 2000 to March 2025. Specifically, we asked two key questions:

- (1) Do equity market movements help forecast U.S. consumer inflation? and
- (2) Does consumer inflation help forecast the DJIA?

Using a time series econometric framework that included unit root testing, Granger causality analysis, Vector Autoregression (VAR), and impulse response functions, we uncovered strong evidence of unidirectional predictability from equity markets to inflation. Granger causality tests confirmed that lagged DJIA values significantly improve the forecast of CPI, while CPI does not significantly Granger-cause DJIA. The VAR(2) model confirmed this asymmetry and showed that a rise in DJIA has a positive and lagged effect on inflation growth.

Impulse Response Functions further illustrated that a one-standard deviation shock to DJIA leads to an increase in CPI growth over the following 2–4 months, reinforcing the notion that stock market performance contains forward-looking information about inflationary pressures. In contrast, CPI shocks had negligible effects on DJIA, supporting the idea that inflation is largely a backward-looking indicator with limited short-term influence on equity prices.

Model diagnostics affirmed that the residuals were well-behaved, with no autocorrelation and limited heteroskedasticity (except in DJIA returns, which is common in financial time series). The VAR model proved dynamically stable and suitable for short-term forecasting. Our out-of-sample forecasts for April 2023 to March 2025 show modest inflation growth and moderately positive, yet volatile, equity performance — broadly consistent with a post-pandemic normalization phase.

These findings have important practical implications. For policymakers, equity market trends could serve as early warning signals of inflation, aiding in more agile monetary responses. For investors, understanding the lead-lag dynamics between DJIA and CPI can inform asset allocation strategies and inflation-hedging techniques. For academics, the results suggest further exploration into nonlinearities and the inclusion of additional financial or macroeconomic variables.

8. References

1. *Federal Reserve Bank of St. Louis. (2025). Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCNS). Retrieved from <https://fred.stlouisfed.org>*
2. *Yahoo Finance. (2025). Dow Jones Industrial Average (DJIA) historical data. Retrieved from <https://finance.yahoo.com>*
3. *Enders, W. (2014). Applied Econometric Time Series (4th ed.). Wiley.*
4. *Stock, J. H., & Watson, M. W. (2019). Introduction to Econometrics (4th ed.). Pearson.*
5. *Hamilton, J. D. (1994). Time Series Analysis. Princeton University Press.*
6. *Lutkepohl, H. (2005). New Introduction to Multiple Time Series Analysis. Springer.*
7. *Sims, C. A. (1980). Macroeconomics and reality. Econometrica, 48(1), 1–48.*
8. *Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. Econometrica, 37(3), 424–438.*

Log CPI: Non-stationary ($p > 0.6$)

Δ Log CPI: Stationary ($p < 0.01$)

A3: VAR(2) Coefficient Estimates